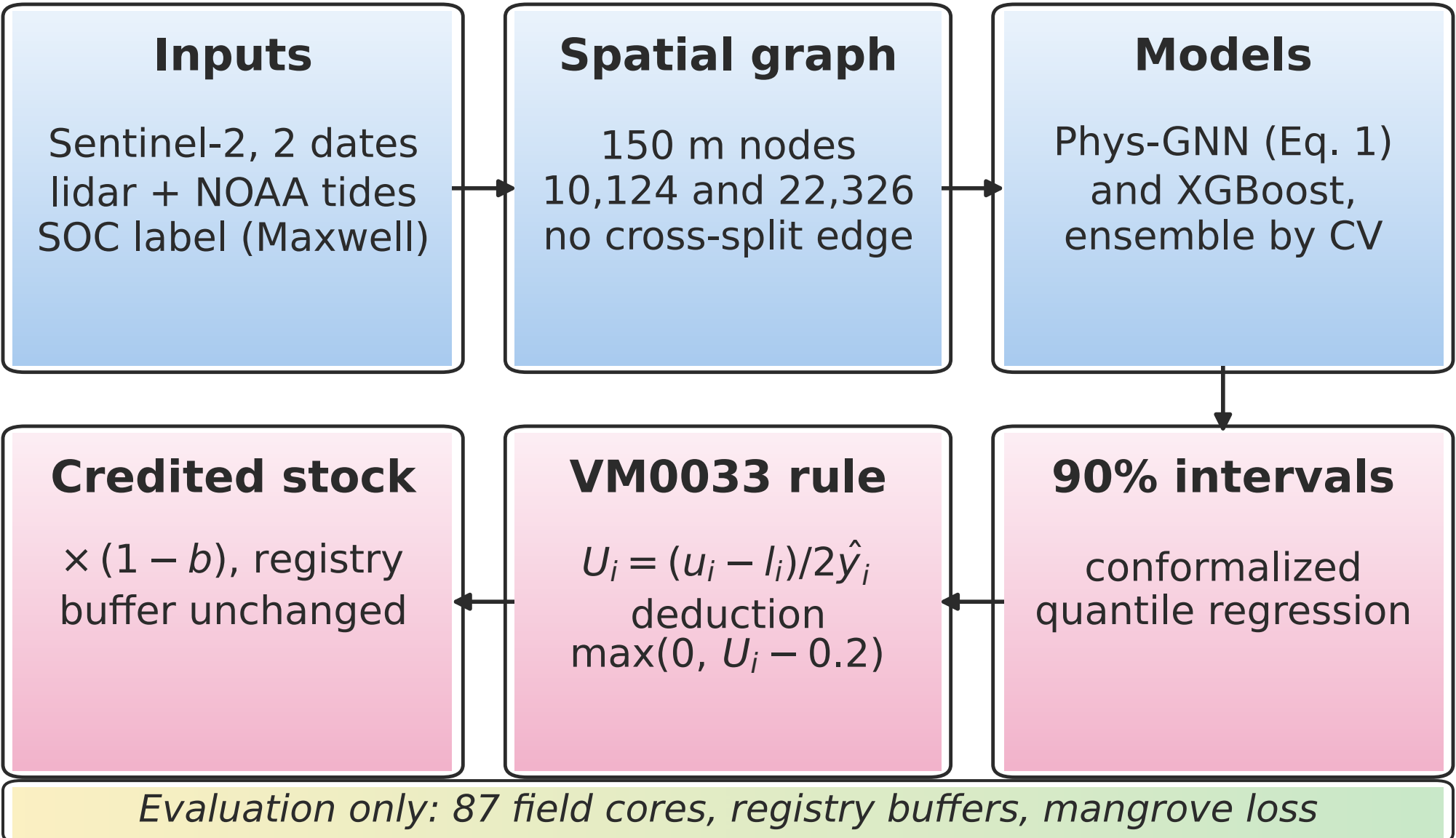

Inputs
Sentinel-2, 2 dates
lidar + NOAA tides
SOC label (Maxwell)
Spatial graph
150 m nodes
10,124 and 22,326
no cross-split edge
Models
Phys-GNN (Eq. 1)
and XGBoost,
ensemble by CV
90% intervals
conformalized
quantile regression
VM0033 rule
$U_i = (u_i - l_i)/2\hat{y}_i$
deduction
max(0, $U_i - 0.2$)
Credited stock
$\times (1 - b)$, registry
buffer unchanged
Evaluation only: 87 field cores, registry buffers, mangrove loss